

KOL2Trade: Learning Trading Strategies from Social Media KOLs via RL

A Unified Pipeline for Data Extraction, Strategy Learning, and
Ranking System

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Motivation

Why Study KOL-driven Trading?

- Financial KOLs strongly influence retail investor behavior.
- KOL content is **asymmetric**:
 - Very explicit on **entries** (“I like stock A”).
 - Almost silent on **holding or exit**.
- Signals are sparse:
 - 40% tickers only appear once.
 - Silence intervals often exceed 100 days.

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Core Problem

How do we turn noisy, incomplete KOL comments into a full trading strategy?

Project Overview

Three-Part Project Structure

1. **Part 1: KOL Data Extraction** YouTube → Transcript → LLM Summary → Training Dataset.
2. **Part 2: Strategy Learning** KOL Baseline + RL (BC + IQL) to learn holding and exit behaviors.
3. **Part 3: Ranking System** A platform where KOLs and LLM/RL agents “compete” with their equity curves.

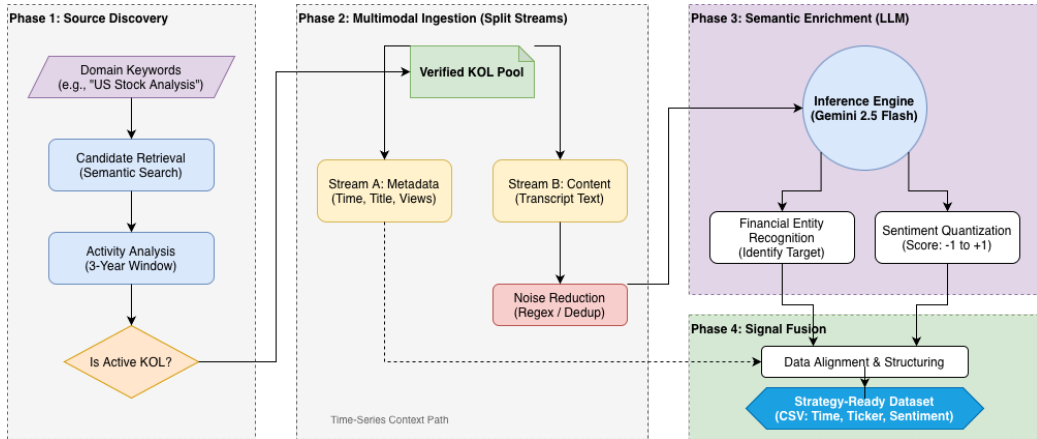
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A unified system from social media signals to actionable trading strategies.

Part 1: KOL Data Extraction Pipeline

End-to-End Pipeline Overview



End-to-End Pipeline Overview (Summary)

Goal: Convert raw YouTube KOL videos into a clean, structured, strategy-ready dataset.

- **Source Discovery** Identify active KOLs using semantic search + 3-year activity analysis.
- **Multimodal Ingestion** Metadata (title, time, views) and transcript text (yt-dlp, cleaned, deduped).
- **Semantic Enrichment (LLM)** Gemini 2.5 Flash extracts companies, summaries, sentiment, and confidence.
- **Signal Fusion** Align all information by (date, ticker) → produce strategy-ready signals.

Output → A clean (Time, Ticker, Sentiment) dataset for RL training (Part 2).

Phase 1 & Phase 2: Discovery and Ingestion

Phase 1 — Source Discovery

- Keyword search (e.g., “US stock analysis”).
- Semantic retrieval of candidate channels.
- 3-year activity analysis using YouTube API.
- Filter to obtain the **Verified KOL Pool**.

Phase 2 — Multimodal Ingestion

- **Stream A (Metadata):** Retrieve title, publish time, views.
- **Stream B (Transcripts):** Download & clean subtitles using yt-dlp + regex deduplication.
- Preserve synchronized time-series context for each video.

Phase 3 & Phase 4: Semantic Enrichment and Signal Fusion

Phase 3 — Semantic Enrichment (LLM)

- Use Gemini 2.5 Flash to extract:
 - financial entities (tickers / companies),
 - supporting excerpts,
 - sentiment score $[-1, +1]$,
 - confidence score $[0, 1]$.

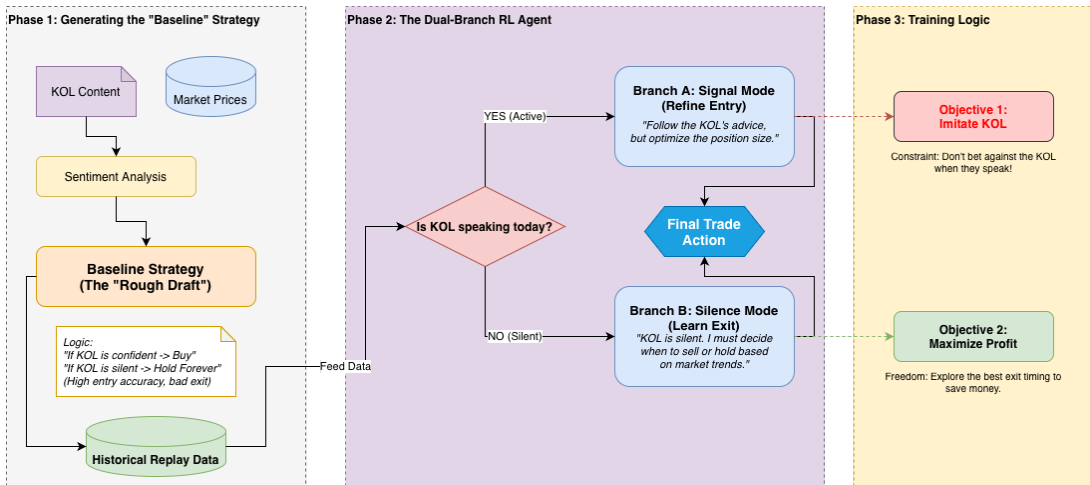
Phase 4 — Signal Fusion

- Merge metadata + transcript insights.
- Align by (date, ticker) into time-series format.
- Produce the **strategy-ready dataset**: (Time, Ticker, Sentiment).

This dataset becomes the sole input to RL training (Part 2).

Part 2: From KOL Data to Trading Strategy

Dual-Branch RL: Core Mechanism Overview



Dual-Branch RL: Key Ideas

- **Phase 1 — Baseline Strategy** Convert KOL sentiment into a rough draft strategy (“follow when they speak, hold when silent”).
- **Phase 2 — Dual-Branch RL Agent**
 - **Branch A: Signal Mode** — Follow KOL direction but optimize sizing.
 - **Branch B: Silence Mode** — Learn exit timing when KOL is silent.
- **Phase 3 — Training Logic** Two distinct objectives: (1) imitate KOL, (2) maximize profit by optimizing exit timing.

Baseline Strategy: Faithful to KOL Intent

- Convert sentiment $\times$ confidence into raw score.
- Normalize into daily portfolio weights.
- Baseline reflects “What would KOL do if they traded daily?”

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Limitation

Baseline cannot decide:

- When to reduce exposure.
- When to exit.
- How to react in long silence periods.

RL Objective

- Learn **holding** and **exit** behaviors during KOL silence.
- Reward = daily portfolio return.
- Inputs:
 - Text embedding (ModernBERT)
 - Market window features
 - Last position
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RL learns everything KOLs do NOT say.

Dual-Branch Policy: Key Innovation

1. Signal Branch (KOL Mentioned)

$$a^{sig} = b(1 + \alpha \tanh(\delta))$$

Adjusts baseline magnitude but not direction.

2. Silent Branch (No KOL Mention)

$$a^{nosig} = last_pos \cdot \sigma(\beta\delta)$$

Learns decay/exit behavior.

3. Combined Policy

$$a = ha^{sig} + (1 - h)a^{nosig}$$

Interpretation

Model learns “how to hold” and “when to sell” independently of KOL noise.

Training: BC + IQL

- **Behavior Cloning:**
 - Aligns signal branch with KOL baseline.
- **IQL (Implicit Q-Learning):**
 - Learns long-term profitable exit rules.
- **Fidelity Penalty:**
 - Prevents over-deviation from KOL intent.

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Result: A strategy that is faithful but smarter than KOLs.

Part 2 Results: Backtests

MarketBeat Backtest Results

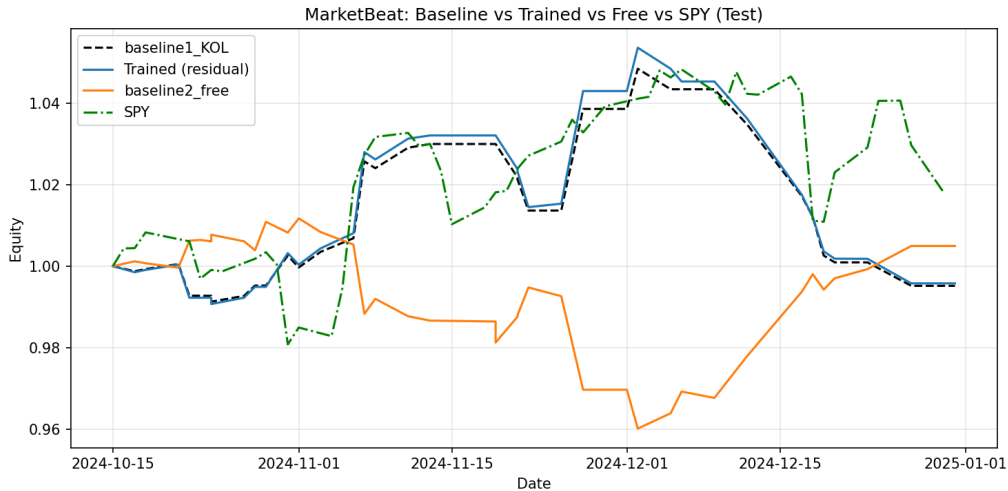
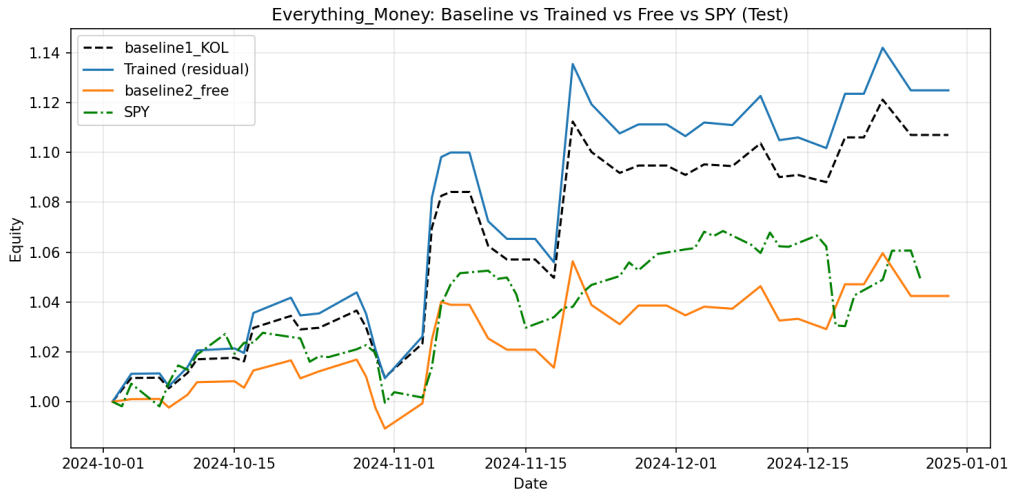


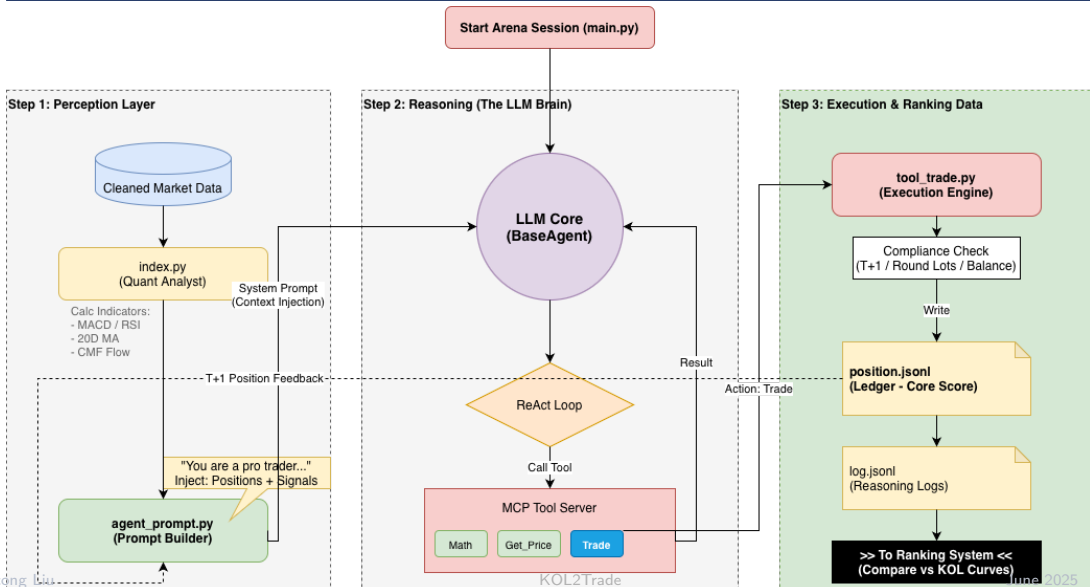
Figure: Baseline vs Trained vs Free vs SPY

Everything Money Backtest Results



Part 3: Strategy Evaluation and Ranking System

Arena Architecture Overview



How the Ranking Curve is Produced

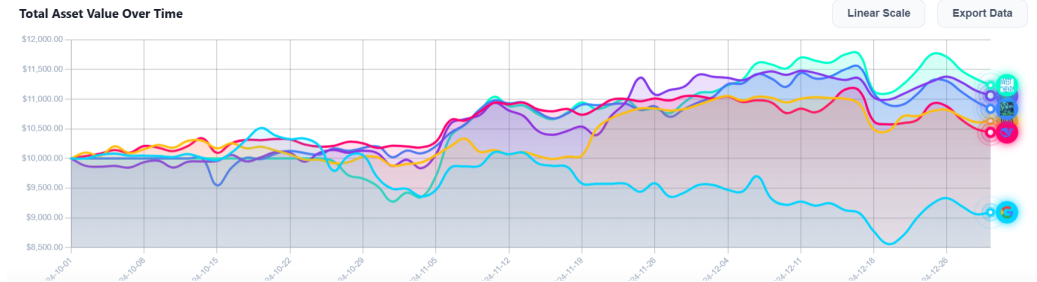
The ranking plot summarizes the entire arena workflow:

- **1. Unified Market Environment** All strategies receive the same anonymized market data (from the ETL pipeline) to ensure fairness.
- **2. Strategy Execution via the Arena** Each agent operates independently:
 - Perception layer computes indicators (MACD/RSI/MA/CMF).
 - LLM or RL agent forms decisions through ReAct reasoning.
 - All orders are executed through the same compliance engine (`tool_trade.py`, enforcing $T+1$ and round lots).
- **3. Equity Curve Construction** After each execution step, the environment records:
 - `position.jsonl` $\rightarrow$ daily portfolio value,
 - `log.jsonl` $\rightarrow$ reasoning trace.

These daily values accumulate to form each curve in the ranking graph.

- **4. Final Leaderboard Computation** Metrics computed from the curves include: total return, Sharpe ratio, and maximum drawdown.

Final Ranking Results (Overall Performance)



- Each curve represents one strategy competing in the unified arena:
 - **KOL baseline signals**
 - **RL-trained residual model**
 - **Free model (unconstrained)**
 - **LLM tool-use agent**
- The arena computes:
 - total return,
 - Sharpe ratio,

Interpretation of Ranking Results

- **RL significantly improves hold/exit behavior** — especially during KOL signal silence periods.
- **KOL baseline preserves intent** but lacks risk control and exit timing.
- **Free model is unstable** because it is unconstrained by KOL semantics.
- **LLM agent performs competitively** due to explicit reasoning and tool-use.

Overall: RL delivers best performance while staying faithful to KOL signals.

Conclusion

Conclusion

- Built a full pipeline from:
 - Social media $\rightarrow$ structured data
 - Structured data $\rightarrow$ RL strategy
 - Strategy $\rightarrow$ public ranking system
- RL learns behaviors KOLs do not specify.
- Ranking platform demonstrates strategy quality transparently.

Future Work

- Expand to multi-KOL ensemble strategies.
- Add risk control modules (vol targeting, stop-loss).
- Deploy ranking dashboard as live website.
- Explore Decision Transformers and multi-agent RL.

Thank you!

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